N S RAJU INSTITUTE OF TECHNOLOGY

(AUTONOMOUS)

(Affiliated to JNTU-GV Vizianagaram)

SONTYAM, ANANDAPURAM, VISAKHAPATNAM – 531 173

Class : III B.Tech I Sem (Common to CSD/CSM)
Regulations : AR23

Academic Year : 2025-26 Max Marks: 70M

Sub: Computer Networks (23CS601)
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Semester V Regular Examination Key And Scheme

Prepared by
Mr.A.Suraj Kumar,
Asst.Prof ,
Dept of CSE(DS),
NS Raju Institute of Technology(A).

Part A Short Answer Questions
5X2=10

a) Describe unguided transmission media with two examples [2M]

Unguided Transmission media– 1M, Two examples– 1M
Unguided media refers to wireless transmission media without a physical cable.
Examples: Radio waves that can be transmitted in all directions (WiFi, Bluetooth etc), Microwaves used for terrestrial communication and Infrared waves used for short distance communication as used in TV remotes.

b) Differentiate between Go-Back-N and Selective Repeat [2M]
Any two differences mainly only damaged frame sent in Selective Repeat– 2M

Feature	Go-Back-N	Selective Repeat
Retransmission Strategy	Any frames sent if believed t o have been	Only the suspected damaged or lost frames

	damaged or lost then all frames are resent from the lost to the last packet transmitted	are retransmitted.
Order of delivery for packets	The Go-Back-N protocol ' s does not accept out-of-order packets and entire window is retransmitted	Out-of-order deliveries are accep ted in Selective Repeat
Window Size	Here the sender window size is N whereas the receiver window size is 1	Here both the sender window and the receiver window sizes are N.

c) What is the purpose of tunneling in networks? [2M]
Tunneling— 1M, Purpose— 1M
Tunneling in computer networks encapsulates one network protocol within another protocol, creating a secure, virtual link over an existing network . Tunneling protocols also ensure that the data is coming from a verified source and hasn't been tampered with in transit, which is key for data integrity.

d) What is congestion? What are the factors that cause congestion? [2M]
Congestion—1M, Mention any two factors mentioned for congestion 1M

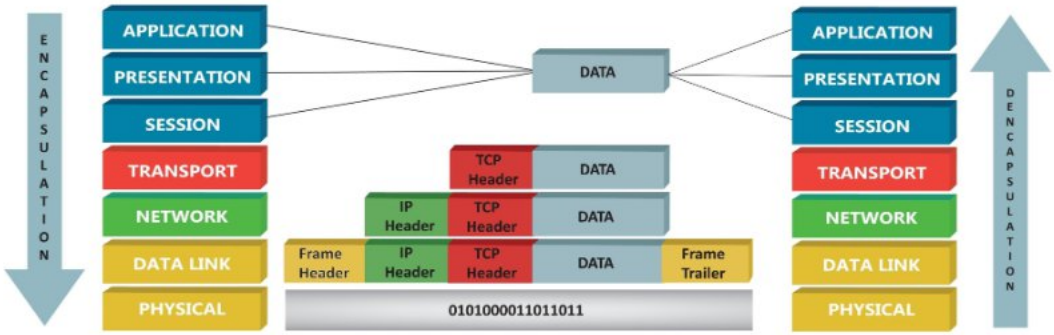
Congestion: It is the reduced quality of service that occurs when a network node or link carries more load than its capacity. Factors that cause congestion are low bandwidth, routers having limited processing power, poor routing decisions, excessive retransmission etc.

e) Expand MIME and SMTP. [2M]
MIME—1M ,SMTP — 1M
MIME stands for Multipurpose Internet Mail Extension, SMTP Simple Mail Transfer Protocol

Part B Long Answer Questions
5X12 = 60

2 (a) Explain how does the OSI layer actually work in transmitting of data from one system to another system? **[12M]**

OSI Layer Diagram—2M, Layer Purpose—5M, Example with each layer purpose justification— 5M



Communication in the OSI Model follows a systematic process called **encapsulation and de capsulation** . Encapsulation is performed at the sender end and decapsulation at the receiver end.

Each of the seven OSI model layers communicates with layers below and above it.
For example, the application layer interacts with software applications, while the presentation layer provides encryption and data compression. The session layer creates communications between devices.
The transport layer breaks data into chunks called segments to send them. Then the receiving device reassembles the segments before the network layer breaks them into smaller packets to send to other networks.
The data link layer facilitates data transfer between devices on the same network, and, finally, the physical layer tran sfers data in ones and zeros .
For example: Sending an email and receiving it
At the sender, data originates at the Application Layer (Layer 7)— when you send an email using an email client. As the data moves down the layers (7 → 1), each layer adds its own header (and sometimes trailer) containing control information needed for that layer 's function:
Layer 7 (Application): Adds user-level data (e.g., email content).
Layer 6 (Presentation): Adds formatting/encryption info (e.g., converts email to MIME format).
Layer 5 (Session): Adds session control info (e.g., establishes/maintains email session).
Layer 4 (Transport): Adds port numbers and sequence info (e.g., TCP header for email delivery reliability).
Layer 3 (Network): Adds logical addressing (e.g., IP header with source/ destination IP).
Layer 2 (Data Link): Adds MAC addresses and a trailer (e.g., Ethernet header + FCS for error detection).
Layer 1 (Physical): Transmits raw bits over the medium (e.g., electrical signals over cable).
This encapsulated data is sent across the network.
At the receiver, the data enters at Layer 1 and moves up the stack (1 → 7). Each

layer strips off (decapsulates) its corresponding header/trailer, processes the info, and passes the remaining data upward:
Layer 1 receives bits.
Layer 2 checks MAC addresses and FCS, removes its header/trailer.
Layer 3 examines IP addresses, removes IP header.
Layer 4 uses port info to deliver to correct application, removes TCP header.
Layers 5 –7 reverse session, presentation, and application processing — finally delivering the original email to the recipient 's email client.
Thus, each layer 's header/trailer enables peer-to-peer communication between the same layers on sender and receiver, ensuring end-to-end data integrity, addressing, error control, and proper interpretation — making seamless communication (like sending/receiving emails) possible across diverse systems.

(OR)

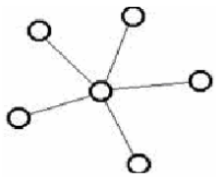
2 (b) i) Explain any three network topologies used in the campus networking. **[6M]**

Star, Ring , Bus, Mesh, Tree or Hybrid Topology diagrams—3M Explanation of topology — 2M

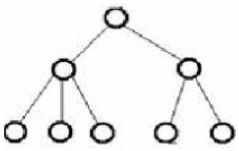
Network topology is the physical and logical arrangement of devices, connections and data circuits in a network. Three network topologies used in campus networking are:

(A) STAR (B) MESH (C) TREE (D) Hybrid

Star: All devices connect to a central hub or switch. Most commonly used topology used in networking.


Star

Tree: Tree topology uses a central root node connected to multiple star-configured subnetworks .


Tree

Tree topology allows organized scalability across departments or buildings.
It has a single point of failure at the root node and its complex structure increases cost and

Mesh: Mesh topology is a network structure where each node is interconne cted with multiple other nodes for reliable data transmission.


Mesh

Mesh is expensive and complex to implement due to the high number of cables and network ports required .
In a full-mesh topology each of the n nodes is connected to every other node requiring n(n-1)/2 physical links.

2 b ii) Compare Guided media and Un Guided media **[6M]**

Guided Media and Unguided media differences any four including definition— 5M

Feature	**Guided Media**	**Un Guided Media**
Definition	Transmission media that use a physical path to carry and direct data signals from one device to another	Transmission media that transport electromagnetic waves without using a physical conductor
Signal Behavior	Signal is confined , it is physically trapped inside the cable and cannot escape.	Signal is broadcast and the signal is radiated outwards in all directions
Physical Barrier	Unaffected by walls or obstacles as the cable is installed around or inside them	Severely affected; signals are blocked by walls and terrains reducing range
Range	High-speed range is fixed and long	High-speed range is variable and short

3 (a) Explain about Cyclic Redundancy Check (CRC) with an example **[12M]**

CRC Definition— 3M, Algorithm— 4M, Example— 5M

CRC, or Cyclic Redundancy Check, is an error-detecting code that appends a short checksum to data, allowing the receiver to verify its integrity .

CRC uses Generator Polynomial which is available on both sender and receiver.

Encoded Data Generation from Generator Polynomial (Sender Side)

1) The binary data is first augmented by adding k-1 zeros in the end of the data.
2) Then, modulo - 2 binary division is used to divide binary data by the key and remainder of division is stored.

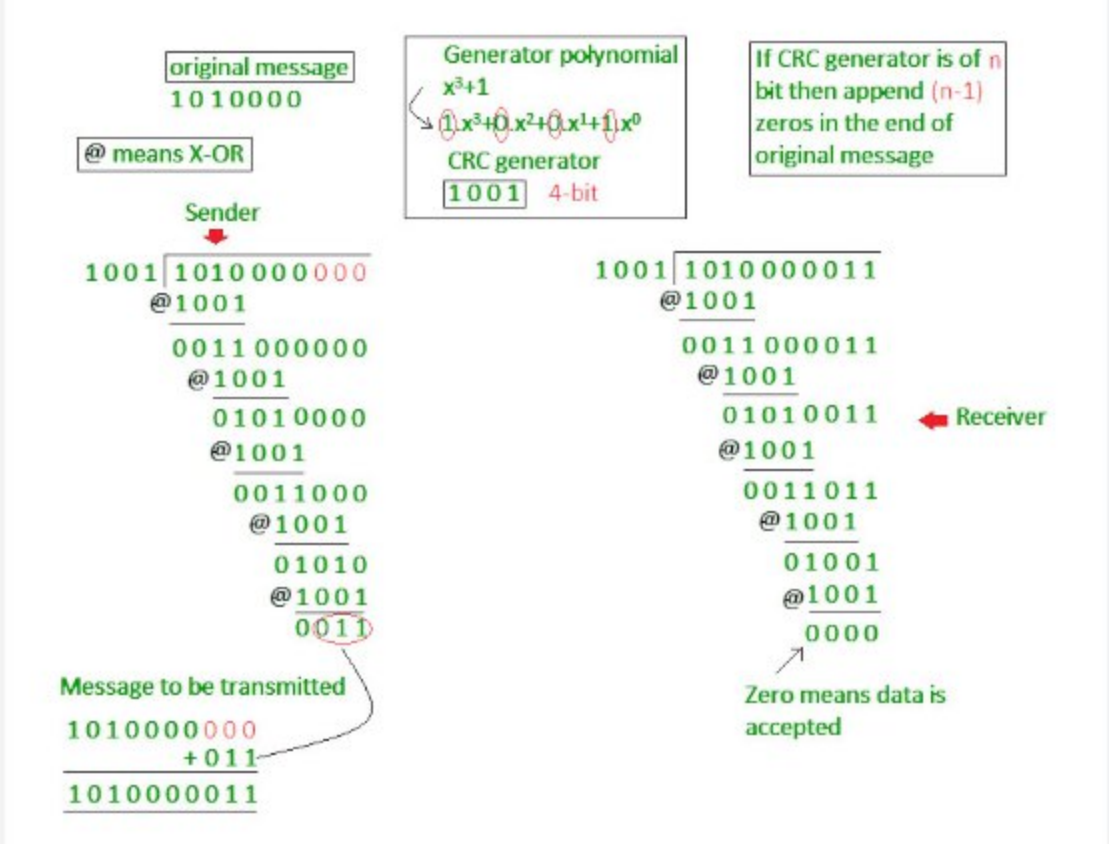
At last the remainder is appended at the end of the data to form the encoded data which is later sent.

Checking Error in Transmission (Receiver Side)

1) After receiving the data, to check if the data is error free, perform the modulo-2 division again.
2) If the remainder is 0, then there are no errors, otherwise, the data is

faulty and contain transmission errors.

Example:



(OR)

3 b i) Explain about Sliding window protocol **[6M]**

Sliding window protocol— 3M Example— 3M

The Sliding Window Protocol is a flow control and error control mechanism used in the Data Link Layer and Transport Layer.

The term "sliding window" comes from the conceptual window of acceptable frames that slides over the sequence numbers as frames are sent and acknowledged.

1. Go-Back-N (GBN)

Mechanism: When a single frame is lost, the receiver discards all subsequent correctly received frames and only acknowledges the last frame received in order.

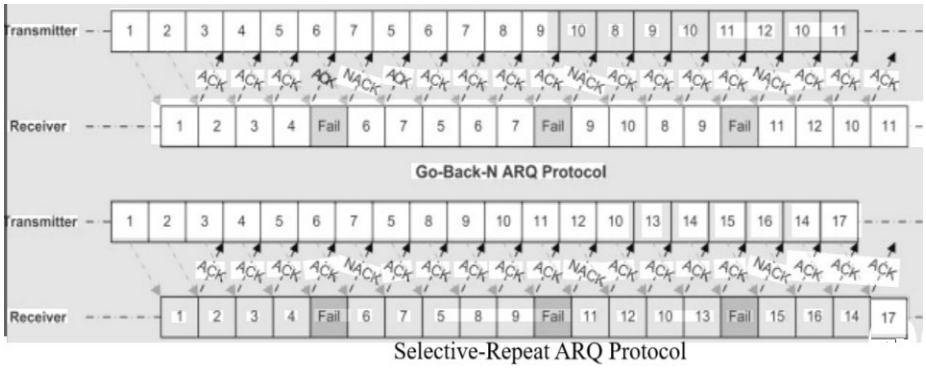
Efficiency & Use: The sender must then retransmit the lost frame and all frames sent after it, making it simple to implement (W_r=1) but less efficient on networks with high latency or frequent errors.

2. Selective-Repeat (SR)

Mechanism: When a frame is lost, the receiver buffers and keeps all subsequent correctly received frames and sends an explicit request only for the missing frame.

Efficiency & Use: The sender retransmits only the single missing frame, making it highly efficient in bandwidth usage but requiring more complex logic & larger buffer sizes at the receiver (W_s 2k-1),

W, 2k-1).



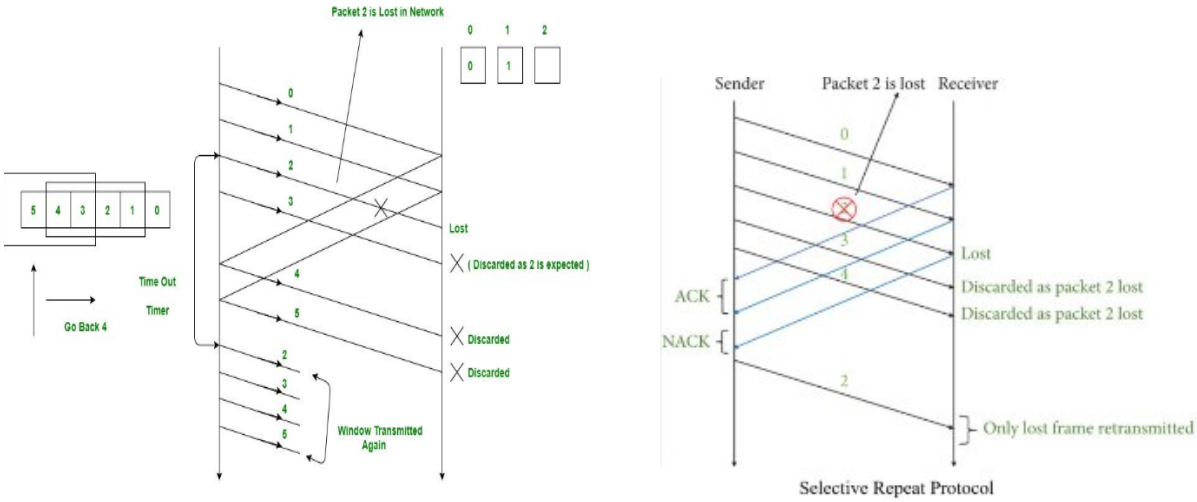


Fig. Go-Back-N (GBN) Protocol

3 b ii) Explain token bus and token ring **[6M]**

Token bus— 3M, Token ring— 3M

Token Bus (IEEE 802.4) is a popular standard for token passing LANs.

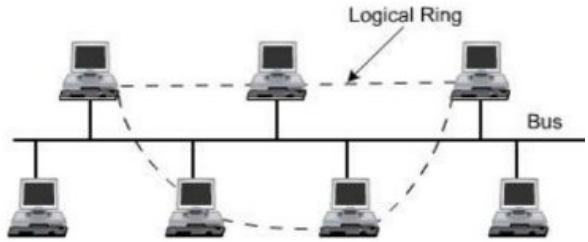


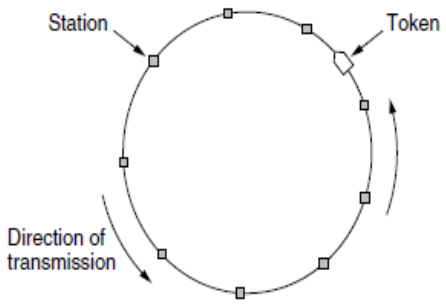
Fig. Token Bus

In a token bus LAN, the physica l media is a bus or a tree. Logically the stations are connected to a ring.

The token is passed from one user

Token Ring: Token Ring (IEEE 802.5)

In a **Token Ring network** , stations are conı



A special frame called a **token** circulates around the ring and gives permission to transmit.

A station can send data **only when it holds the token** ; after sending, it passes the token to the next station.

Frames travel in **same direction as token** & are removed from the ring by sender or

Fig. Token Ring

4 a) i) Describe the hierarchical routing algorithm and illustrate it with an example. **[6M]**

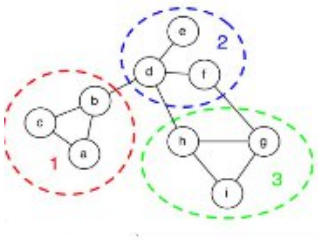
Hierarchical routing— 3M, Example of hierarchical routing— 3M

Hierarchical routing is a strategy designed to make routing more scalable in very large networks by reducing the size of the routing tables and the volume of routing update traffic.

As the network grows, the routing tables become large, making lookups slow and consuming excessive memory and bandwidth for updates.

Principle: Routers are divided into regions.

Example:



All distances in the figure are 1 hop

The goal is to generate the routing table for **Router a** in Region 1.

Destination	**Line**	**Hops**
a	--	0
b	b	1
c	c	1
d	b	2
e	b	3
f	b	3
g	b	3
h	b	4
i	b	4

Destination	**Line**	**Hops**
A	--	0
B	b	1
C	c	1
Region 2 (d,e,f)	b	2
Region 3 (g,h,i)	b	3

(a) Full Table for **1a** (b) Hierarchical Table for **1a**

Hierarchical routing has reduced the entries from 9 to 5.

4 a ii) Differentiate between closed loop and open loop congestion control mechanisms. [6M]

Open Loop Vs Closed Loop Congestion control— 6M

Feature	Open Loop	Closed Loop
Definition	Prevents congestion before it occurs	Addresses congestion after it has occurred
Feedback	No feedback is used; control action not dependent on output .	Uses feedback to adjust traffic flow
Mechanism	Relies on prevention policies using retransmission timers or admission control	Relies on detection and reaction methods- choke packet, backpressure and implicit signaling
Simple Action	The network sets strict entrance rules for traffic to prevent too much data from getting onto the highway in the first place.	When congestion is detected , it sends an immediate warning signal back to the data source, forcing the sender to slow down or stop sending temporarily.

(OR)

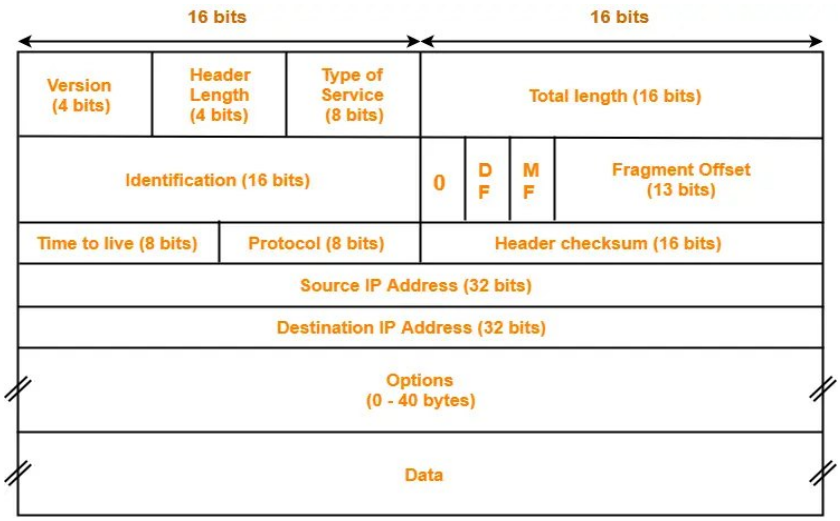
4 b i) Illustrate the IPv4 header format and explain the significance of all the fields. **[6M]**

IPv4 header format— 3M, All fields –mainly any 4 fields— 3M

Internet Protocol Version 4-

IPv4 short for Internet Protocol Version 4 is the fourth version of the internet protocol (IP).

- IP is responsible to deliver data packets from the source host to the destination host.
- This delivery is solely based on the ip address in the packet headers.
- IPv4 is a connectionless protocol for use on packet switched networks



IPv4 Header

1. Version- It is a 4 bit field that indicates the IP version used.

2. Header Length-

- Header length is a 4 bit field that contains the length of the IP header.
- It helps in knowing from where the **actual data begins.**

Type of service: Low Delay, High Throughput, Reliability (8 bits)

Total l ength (16 bits) : Specifies the size of the packet. This allows the receiving host to know how much data to expect and when the packet ends.

Identification: Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)

Flags: 3 flags of 1 bit each : Reserved bit (must be zero), (DF)do not fragment flag, (MF) more fragments flag

Fragment Offset: Represents the number of Data Bytes ahead of the particular fragment in the particular Datagram. Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes

Time to Live (TTL): The 8-bit TTL (IPv4) or Hop Limit (IPv6) field prevents packets from endlessly circulating the network in case of routing loops. It is decremented by each router it passes through, and when the value reaches zero, the packet is discarded

Protocol : Protocol field identifies the protocol of the payload, allowing the receiving host to pass the data to the correct higher-layer protocol, such as TCP, UDP, or ICMP.

Checksum: To check errors in the datagram which is 16 bits.

Source Address : The source address field contains the IP address of the device that sent the packet. **Destination Address :** This 32-bit field specifies the IP address of the intended recipient.

Options : It can be used for network testing, debugging, or security, though it is rarely used in practice due to performance overhead.

4 b ii) What are key differences between IPv4 and IPv6? Discuss their advantages and limitations **[6M]**

Key differences between IPv4 and IPv6—3M, One advantage—2M, One limitation of IPv4, IPv6— 1M

Feature	IPv4	IPv6
Addressing Scheme	Originally used classful addressing (A ,B,C) leading to massive wastage, now using CIDR to manage scarcity of addresses	Mandatory CIDR from inception. The scheme is designed to be classless simplifying routing table entries
Addressing Length	32 bits divided into 4 octants **192.168.10.134**	128 bits divided into 8 groups separated by colons. 2001:0db8:85a3:0000:0000:8a2e:0370:7334
Fragmentation	Routers handle it consuming resources	Handled by only the sending host improving router efficiency
Checksum	Includes a Header Checksum, requiring every router to recalculate and verify it, slowing down processing.	Eliminates checksum field speeding up router processing since integrity handled by other layers
Advantage	Universally compatible and used by almost all networking devices to communicate	Vast Address Space that eliminates the need for NAT
Limitation	Address scarcity which forced to use Network Address Translation (NAT) to cope up.	The IPv6 header is always 40 bytes, which creates unnecessary overhead when sending tiny amounts of data compared to the minimum 20-byte header of IPv4.

5 a Demonstrate how connection management is done in TCP **[12M]**

Connection management in TCP— 12M

TCP Connection Management : TCP uses a finite-state machine with different states to manage how a connection is created, used, and closed.
Every TCP connection starts in CLOSED state.

1. Connection Establishment (Three-Way Handshake)

There are two sides: Client and Server.
Client Side (Active Open)
Client wants to connect → sends SYN , State becomes SYN-SENT
Server replies with SYN + ACK
Client sends ACK back , State becomes ESTABLISHED
Now data can be sent in both directions.
Server Side (Passive Open)
Server executes LISTEN → waits for requests , State becomes LISTEN
When SYN from client arrives → server sends SYN + ACK , State becomes SYN-RCVD
When it receives the final ACK →
State becomes ESTABLISHED
Now communication begins.

2. Connection Termination

Closing a TCP connection happens in two steps, because TCP is full-duplex
Client Closing (Active Close)
Client sends FIN , → State: FIN-WAIT-1
Server sends ACK , → State: FIN-WAIT-2
Later, server also sends FIN
→ Client sends ACK
→ State: TIME-WAIT
Client waits for a short time (to ensure all packets die off) → Then goes to CLOSED
Server Closing (Passive Close)
Server receives FIN , → State: CLOSE-WAIT
When the server application is done, it also sends FIN , → State: LAST-ACK
When the client sends the final ACK , → Server goes to CLOSED

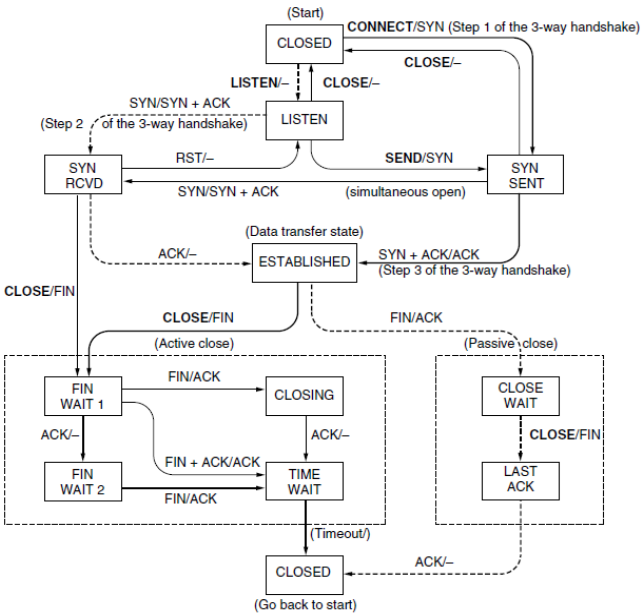


Figure TCP connection management finite state machine. The heavy solid line is the normal path for a client. The heavy dashed line is the normal path for a server. The light lines are unusual events. Each transition is labeled with the event causing it and the action resulting from it, separated by a slash.

(OR)

5 b i) Explain the three-way handshake mechanism used in TCP connection establishment. **[6M]**

Three-way handshake diagram—3M Expla nation— 3M

TCP Connection Establishment (3-Way Handshake) : TCP is a reliable, connection-oriented protocol. Before any data is exchanged, TCP uses a three-way handshake for establishing connection.

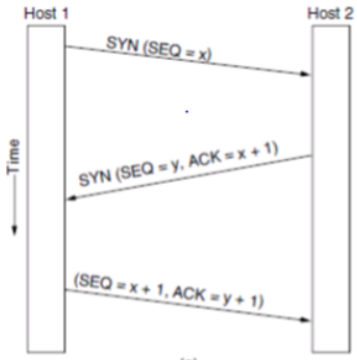


Fig. Connection Establishment

TCP Connection Establishment:
Step 1: The initiating client requests a client-to-server communication session with the server.
Step 2: The server acknowledges the client-to-server communication session and requests a server-to-client

5 b ii) Explain Remote Procedure Ca

Remote Procedure Call (RPC) Diagram—2M and its purpose— 4M

Remote Procedure Call (RPC) : Remote Procedure Call (RPC) is a protocol that enables a client to run a procedure on a s erver as if it were local.

It works by making a remote procedure call appear like a local one to the developer, abstracting the underlying details of inter-process communication across the network. The client calls a procedure, which sends the request to the server, where the procedure is executed and the result is sent

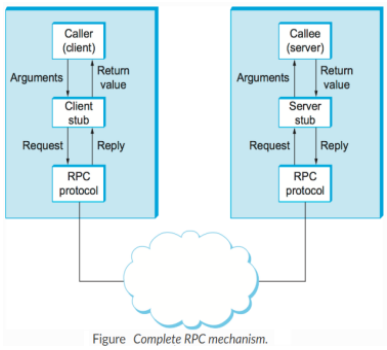


Figure *Complete RPC mechanism.*

6 a i) Explain the resource records of DNS with its attributes [6M]

Resource Record definition its five tuple description— 4M Purpose of DNS — 2M

Resource Record: A DNS resource record is a type of data stored in the Domain Name System (DNS) that contains information about a domain name, such as its IP address, mail servers, and other details. Its purpose is to map human-readable domain names to the corresponding network addresses .

Format of a resource record is a 5 tuple: **Domain_Name Time_To_Live Class Type Value**

Tuple	**Meaning**
Domain_Name	Identifies the specific domain or name that the record applies to, serving as the primary search key for queries.
Time_To_Live	Indicates the stability and freshness of the record's information
Class	Specifies the protocol family for the record, which is almost always IN for Internet information.
Type	Defines the kind of resource information contained in the record (e.g., A, CNAME, MX).
Value	Holds the data of the record, such as an IP address, domain name, or text string, depending on the record type.

6 a ii) Explain SMTP [6M]

SMTP definition—1M and its role in sending mail –5M

Simple Mail Transfer Protocol (SMTP) is the application-layer protocol used to push (send) and transfer email across the Internet. It functions as the core language mail servers use to exchange messages.

ASCII-Based: SMTP is fundamentally an ASCII-based protocol, meaning it was originally designed only to handle plain text characters (7-bit ASCII).

Process: It uses a text-based, command-response sequence (e.g., MAIL FROM:, RCPT TO:, DATA) over a TCP connection (typically Port 25 or Port 587).

Function: SMTP is inherently a push protocol because the sender's mail server (client) initiates the connection and actively pushes the message to the recipient's mail server (server).

Role: It is used both by your email client to submit a message to your outgoing server and by the outgoing server to transfer the message to the recipient's incoming server.

Limitation: The text limitation means native SMTP cannot handle multimedia messages (like attachments, images, or non-English characters) directly. So it uses MIME.

As a push-only protocol, SMTP is used only for sending; other protocols (like POP3 or IMAP) are needed for message retrieval.

(OR)

6 b) Explain the architecture of Email **[12M]**

Architecture of email—4M, Explain User Agent, Message Transfer Agent and Final Delivery— 8M

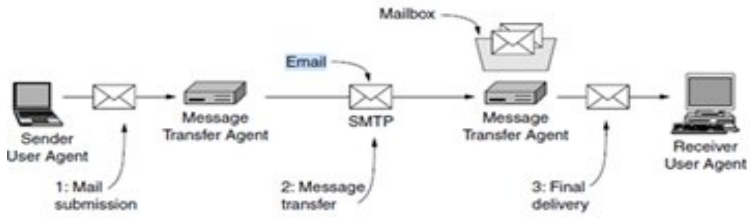


Figure . Architecture of the email system.

Email Architecture:

The email system works in three steps, involving three main components .

(1) User Agent (UA), (2) Mail Transfer Agent (MTA) and (3) Final Delivery.

1. User Agent (UA) : The UA is the software interface (client) used by the sender and receiver (gmail).

Step 1 (Mail Submission): **Composes** the message and **pushes** it into the mail system (server).

Step 3 (Final Delivery): **Pulls** the message from the mailbox for the user to **read** (using POP3/IMAP).

2. Mail Transfer Agent (MTA): It is the server that handles the transmission and storage of mail.

Step 2 (Message Transfer): Uses **SMTP** and **DNS (MX records)** to **reliably relay** the message to the correct destination server.

Step 3 (Final Delivery): **Stores** the incoming message in the recipient's **mailbox** until UA retrieves it.

3. Final Delivery: It is the final phase that makes the message available to the recipient.

Key Difference: It **cannot use SMTP** (which is push-based) because the user may not be online.

Action: The **UA pulls** the mail from the MTA's mailbox using separate protocols like **POP3** or **IMAP** .

Services of Email:

Composition and Addressing : Email allows users to create messages easily and format the necessary recipient and header information.

Transfer and Delivery: The system reliably moves the message from the sender's computer to the recipient's server using SMTP .

Mailbox Management: Users can access, organize, and store their received and sent messages on the server until they are ready to be read.